ECO 110 Assignment # 2
Professor M. McKay Due: November 1, 2026

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*Be as detailed as possible when asked to explain, why, or discuss. The document you submit should be an MS Word document or a PDF. Not a set of image files.

1. **[30pts]** Use the following supply and demand conditions for a Samsung Tablet.

$$QD = 1200 - P \qquad QS = -400 + P$$

a. Solve for the equilibrium price and equilibrium quantity (show your work). (3pts)

$1200 - P = -400 + P$

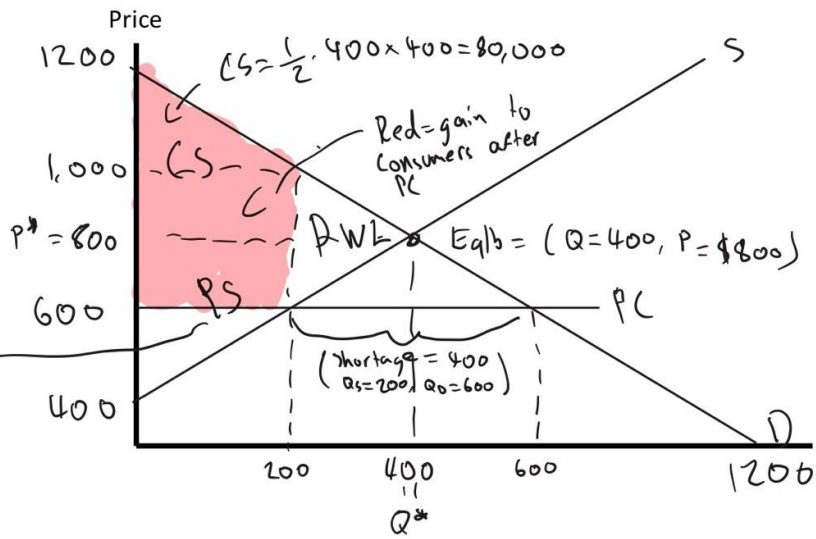
$\frac{1600}{2} = \frac{2P}{2}$

$800 = P$

$Q_D = 1200 - 800 = 400$

$Q_S = -400 + 800 = 400$

Market for Samsung tablets



$PS = \frac{1}{2} \times 400 \times 400 = 80,000$

$Q_D = 1200 - P = 0$

$P = 1200$

$Q_D = 1200 - 0$

$Q_D = 1200$

$Q_S = -400 + P = 0$

$P = 400$

b. Illustrate the two curves with a **fully labeled and accurate graph**. (5pts)

c. At what price does demand no longer exist? (4pts)
Demand disappears at P = $1,200

d. If tablets were free, how many would people take? (2pts)
If tablets were free, people would take 1,200.

e. Calculate the consumer and producer surpluses? (4pts)
CS = 80,000, and PS = 80,000

f. If the government wanted to encourage the use of tablets in schools, would it install a price floor or a price ceiling? **Why?** Between which two prices would the floor or ceiling be placed? **Explain why**. (5pts)

A price ceiling, because it lowers the price so tablets are cheaper for schools It goes between $400 and $800: below $800 so it has an effect, and above $400 so producers still supply tablets.

g. Who would benefit (consumers or producers) from the intervention you chose in part f? **Explain** why that side of the market benefits and **show** on your graph how this benefit is illustrated. (5pts)

Consumers gain from the ceiling, because they pay a lower price ($600 instead of $800). On the graph, the red area shows the CS rising from $80,000 to $100,000. There is also a DWL.

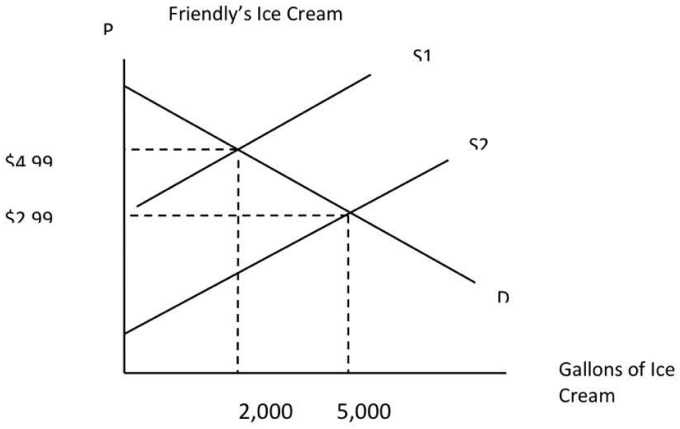
2. **[10pts]** Subway decides to lower the price of its "Foot Long" sub from $7.50 to $5.00. When the price is decreased sales of "Foot Long" subs increase from 2.3 million per day to 4.2 million per day. Assuming all changes sales were due to this price change, what is the price elasticity of demand for Subway "Foot Long" subs?

What does your answer say about the subs and their relationship to other products?

Elasticity is $E_d = 1.46$
$\%\Delta Q = (4.2-2.3)/3.25 = 58.46\%$
$\%\Delta P = |5.00-7.50|/6.25 = 40\%$
$E_d = 58.46/40 = 1.46 > 1 \rightarrow$ elastic

Subs have many substitutes (other sandwich and fast food places), so customers are price-sensetive and switch easily when the price changes.

3. **[20pts]** Use the graph to answer the following questions



a. Do you believe that Friendly's Ice Cream is an elastic or inelastic product? **Explain why**? (4pts). Elastic. Ice cream is a treat, not a necessity, and there are many substitutes for example Ben and Jerry. The Demand curve is relatively flat so a price drop ($4.99 → $2.99) leads to a large increase in quantity (2,000 → 5,000)

b. Prove mathematically, that the ice cream is elastic or inelastic. (2pts)
Demand is elastic, with $E_d = 1.71 > 1 \rightarrow$ elastic.
$\%\Delta Q = (5,000-2,000)/3500 = 85.7\%$
$\%\Delta P = |2.99-4.99|/3.99 = 50.1\%$
$E_d = \frac{85.7}{50.1} = 1.71 > 1 \rightarrow$ elastic.

c. Mathematically prove which S curve the company would prefer to operate on. **Explain why** (based on elasticity) it makes economic sense for this to be true. (8pts)

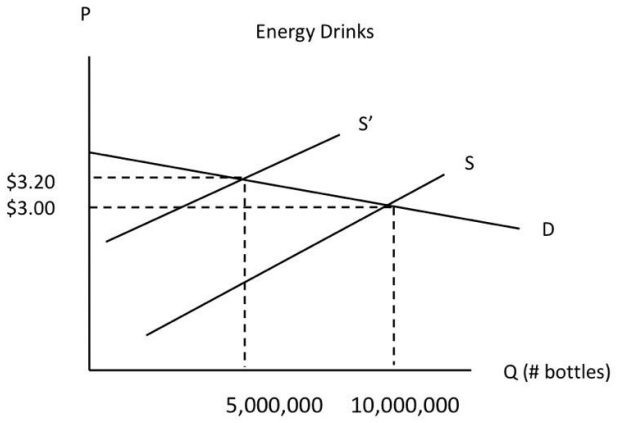
$TR = P \times Q$
S1: $\$4.99 \times 2,000 = \$9,980$
S2: $\$2.99 \times 5,000 = \$14,950$

The company prefers S2 because total revenue is higher. Demand is elastic, so the % increase in quantity (85.7%) is bigger than the % decrease in price (50.1%). A lower price therefore raises total revenue.

d. What would happen to Friendly's demand curve if Ben & Jerry's, Haagen Dazs, and Breyers (all ice cream manufacturers) went out of business? Would Friendly's then decide to increase or decrease their price? **Why?** (6pts)

Friendly's demand curve shifts right (more customers) and becomes steeper/more inelastic (fewer substitutes). Friendly's would increase its price, because when demand is inelastic, a higher price raises total revenue.

4. **[20pts]** Use the graph to answer the following questions.



Event: The government has decided that energy drinks are unhealthy and are charging producers a $1.00 per bottle tax on the production of energy drinks.

a. Why has the production tax shifted the supply curve from S to S'? (2pts)

The tax is a $1 cost per bottle for producers, so they need $ more to supply the same quantity. Supply shifts left from S to S'.

b. Calculate the price elasticity of demand for energy drinks. (2pts)

%ΔQ = (15M − 10M) / 7.5M = 66.7%
%ΔP = (3.20 − 3.00) / 3.10 = 6.45%
E_d = 66.7 / 6.45 = 10.33 > 1 → Very elastic

c. How much of the $1.00 tax is paid by the consumers? (2pts)

$3.20 − $3.00 = $0.20

d. How much of the $1.00 tax is paid by the producers? (2pts)

$1.00 − $0.20 = $0.80

e. How much tax revenue does the government earn from this tax? (2pts)

$1.00 × 5,000,000 = $5,000,000

f. How much of the tax revenue comes from consumers? (2pts)

$0.20 × 5,000,000 = $1,000,000

g. How much of the tax revenue comes from producers? (2pts)

$0.80 × 5,000,000 = $4,000,000

h. In this scenario, **explain** why the producers are paying the larger part of the tax burden. (6pts)

Demand is very elastic (E_d = 10.33) because consumers have a lot of substitutes (coffee, soda, tea) and energy drinks are not a necessity. If producers raised the price by the full $1 consumers would just stop buying. So producers can only raise the price by $0.20 and have to absorb the remaining $0.80 themselves. The side of the market that is less elastic (producers) pays the bigger share of the tax.